Attentional Feature Refinement and Alignment Network for Aircraft Detection in SAR Imagery

Yan Zhao, Lingjun Zhao, Zhong Liu, Dewen Hu, *Senior Member, IEEE*, Gangyao Kuang, and Li Liu, *Senior Member, IEEE*

***Abstract*— Aircraft detection in synthetic aperture radar (SAR) imagery is a challenging task in SAR automatic target recognition (SAR ATR) areas due to aircraft's extremely discrete appearance, obvious intraclass variation, small size, and serious background's interference. In this article, a single shot detector (SSD), namely, attentional feature refinement and alignment network (AFRAN), is proposed for detecting aircraft in SAR images with competitive accuracy and speed. Specifically, three significant components, including attention feature fusion module (AFFM), deformable lateral connection module (DLCM), and anchor-guided detection module (ADM), are carefully designed in our method for refining and aligning informative characteristics of aircraft. To represent the characteristics of aircraft with less interference, low-level textural and high-level semantic features of aircraft are fused and refined in AFFM thoroughly. The alignment between aircraft's discrete backscatting points and convolutional sampling spots is promoted in DLCM. Eventually, the locations of aircraft are predicted precisely in ADM based on aligned features revised by refined anchors. To evaluate the performance of our method, a self-built SAR aircraft sliced dataset and a large scene SAR image are collected. Extensive quantitative and qualitative experiments with detailed analysis illustrate the effectiveness of the three proposed components. Furthermore, the topmost detection accuracy and competitive speed are achieved by our method compared with other domain-specific methods, e.g., dense attention pyramid network (DAPN) and pyramid attention dilated network (PADN), and general convolutional neural network (CNN)-based methods, e.g., Feature Pyramid Network (FPN), Cascade R-CNN, SSD, RefineDet, and RepPoints Detector (RPDet).**

***Index Terms*— Aircraft detection, anchor-guided detection module (ADM), attention feature fusion module (AFFM), attentional feature refinement and alignment network (AFRAN), deformable lateral connection module (DLCM), synthetic aperture radar (SAR), SAR images.**

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I. Introduction

SYNTHETIC aperture radar (SAR) has attractive imaging capabilities in nearly all weather and illumination conditions and SAR image interpretation has numerous applications in civil and military fields, including surface monitoring [1]–[3], transportation management [4], [5], and military surveillance [6], [7]. As a fundamental problem in SAR image interpretation fields, target detection aims at accurately locating and identifying targets of interest from SAR imagery and has been researched for several decades [8]–[11]. As one of the key problems in SAR target detection, aircraft detection has potential applications in modern airport management, military reconnaissance, and so on and has become an independent research direction. Especially with the rapid development of SAR imaging technique, high-resolution SAR images can be accessed more easily than before, enabling even more research opportunities for challenging fine-grained SAR target recognition tasks, including fine-grained aircraft recognition.

The problem of SAR aircraft detection has the following main challenges.

1) *Extremely Discrete Appearance:* Due to the smooth characteristic of aircraft's surface, the appearance of aircraft in SAR images is discrete compared to those in optical images, leading to the lack of appearance information such as geometry and contour cues [see aircraft marked by the real and dotted yellow rectangles in Fig. 1(a) and (b)]. Textural information and backscattering points are important visual cues for SAR aircraft detection.
2) *Large Intraclass Variation:* Variations in aircraft's posture, scale, category, and SAR imaging parameter cause significant impact on the appearance of aircraft in SAR imagery [see Fig. 1(c)–(e) for examples], leading to large intraclass variation. Thus, detecting aircraft with large intraclass variation calls for a precise representation of their salient and constant backscattering information.
3) *Small Targets and Large Scene:* Different from object detection from natural images, e.g., ImageNet [12] and Microsoft COCO [13], SAR images usually cover very large ranges, especially those imaged by satellites and aircraft are with small size and parked sparsely (see Fig. 11 for example), leading to very large searching spaces. Therefore, how to explore context

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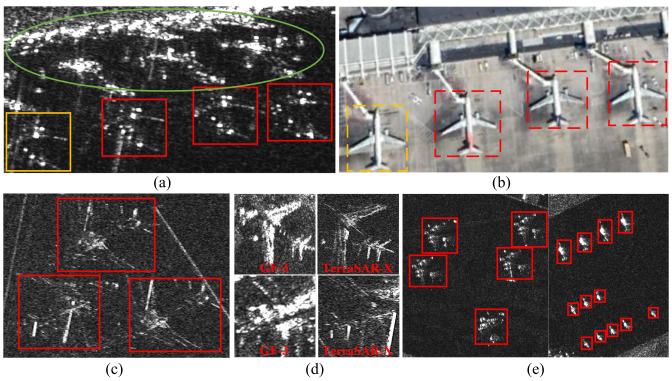

Fig. 1. Aircraft in SAR and optical images. (a) Aircraft in an SAR image. (b) Aircraft in an optical image. (c) Posture sensitivity. (d) Imaging parameter diversity. (e) Scale and class diversity.

information to locate regions of interest (ROI) to increase detection speed plays a key role.

4) *Uninformative Background Clutter:* As shown by the green ellipse of Fig. 1(a), complicated background clutters such as the scattering clusters of buildings are uninformative and could lead to a number of false alarm aircraft. Therefore, suppressing the disturbance of background clutters is important for accurate aircraft detection.

Earlier methods for SAR aircraft detection [14]–[16] rely on handcrafted grayscale features and expert interpretation knowledge of aircraft, e.g., gradients, structures, and contours, and are sensitive to model parameters and lack of generalization to diversified situations, which results in suboptimal detection performance. Recently, convolutional neural networks (CNNs) have achieved remarkable success for object detection in optical imagery [17], [18] and also provide great opportunities for aircraft detection in SAR imagery [14], [19]–[22].

However, most of the current CNN-based aircraft detection methods are directly borrowed from the field of object detection in optical imagery [23]–[26], without fully taking into consideration the aforementioned domain knowledge and challenges of SAR aircraft, that is to say, such detectors are not tailored well for aircraft detection in SAR images, leading to poor interpretability and suboptimal performance. Specifically, aircraft's low-level textural details are not fully considered as the high-level semantic features. Also, the ability for suppressing interference of surroundings is unsatisfied, which leads to insufficient and suboptimal representation for aircraft in SAR images. Besides, it is inappropriate for capturing aircraft's discrete and irregular backscattering points by a traditional convolution, of which the convolutional kernels are inflexible. In addition, detecting sparsely parked small aircraft in large scene images calls for a carefully designed framework with sophisticated detection heads to maintain a tradeoff between accuracy and speed and, however, is neglected in current methods.

In response to the aforementioned challenges, a single shot detector (SSD), namely, attentional feature refinement and alignment network (AFRAN), is proposed for detecting aircraft in SAR images with balanced detection accuracy and speed. With RefineDet [27], an excellent CNN-based detector for handling diversified natural objects, as a basic prototype, the novelty of our method stands on a careful consideration of unique characteristics of aircraft, e.g., structures, appearance, and textures, in SAR images, and is realized by specially designing three crucial modules in AFRAN for feature refinement and alignment. For feature refinement, attention feature fusion module (AFFM), which consists of several feature aggregation pathways followed by a split-attention (SA) block, is designed for representing both textural and semantic features as well as highlighting significant information of aircraft adaptively. For feature alignment, deformable lateral connection module (DLCM) applied at lateral connections of a three-layer fine-grained feature pyramid focuses on aligning discrete backscattering points of aircraft to spots of convolutional kernels with less interference involved. Furthermore, an anchor-guided detection module (ADM) is attached at multiscale feature maps to align sampling points of convolutional kernels with features at unique areas indicated by revised anchors. Our method is evaluated on a self-built SAR aircraft sliced dataset and a large scene image. Extensive quantitative and qualitative experiments illustrate the contributions of the three proposed submodules and the comprehensive performance of our method compared with other CNN-based detectors. In addition, several key factors within our submodules are also discussed in detail.

The contributions of this article are summarized as follows.

1) An SSD AFRAN is proposed in this article for aircraft detection in SAR images by carefully taking the characteristics of aircraft into consideration.
2) To identify and locate aircraft accurately, three significant components, i.e., AFFM, DLCM, and ADM, are designed for refining low-level textural details and high-level semantic features as well as aligning discrete information of aircraft progressively.
3) The topmost detection accuracy with competitive speed is achieved by our method on a self-built SAR aircraft sliced dataset and a large scene SAR image compared with other CNN-based methods, in which the domain knowledge of aircraft is underutilized.

The remainder of this article is organized as follows. In Section II, the related works for aircraft detection in SAR images are provided. In Section III, motivations and our method are introduced in detail. Experiments and analysis are given in Section IV. Finally, Section V concludes the work of this article.

II. Related Works

A. SAR Automatic Target Recognition

SAR automatic target recognition (SAR ATR) generally refers to detect and recognize target signatures from SAR data [28]–[30]. A standard architecture of SAR ATR proposed by the Massachusetts Institute of Technology (MIT) Lincoln Laboratory [31] is shown in Fig. 2, which contains three progressive stages, including detection, discrimination, and classification. Among the three steps, target detection acting as a footstone of the downstream tasks aims at coarsely figuring out potential ROIs (targets) and eliminates background areas.

Fig. 2. Processing flow of SAR ATR.

A discriminator is adopted to identify nonobjects and objects from outputs of the detector. Finally, a classifier is constructed to identify the classes of targets by using diversified domain-specific knowledge [32]–[34].

B. Aircraft Detection in SAR Images

Currently, methods for aircraft detection in SAR images could be coarsely divided into knowledge- and CNN-based paradigms.

1) Knowledge-Based Methods: In these ways, apron areas are first located by statistic models based on grayscale features of backgrounds. Then, potential aircraft within these regions is identified by employing various handcrafted features according to prior knowledge of aircraft, e.g., shapes, contours, and attributes of backscattering points. Among these methods, Hu *et al.* [16] introduced a generalized gamma mixture distribution (GGMD)-based detector to detect aircraft in nonhomogeneous backgrounds. Dou *et al.* [14] proposed a similarity measurement to identify candidates by using the Kullback–Leibler divergence (KLD) of Gaussian mixture model (GMM) based on saliency maps and scattering structure features of aircraft. Besides, Zhang *et al.* [15] introduced an active shape model (ASM) for contour evaluation by utilizing geometric information of aircraft. Needless to say, aircraft could be detected by designing complicated statistic models and combining various types of knowledge of aircraft together. However, the detection performance and generalization of these methods usually degenerate drastically where obvious mismatches exist between inflexible handcrafted features and diversified appearance of aircraft.

2) CNN-Based Methods: The CNN-based object detectors could be finely divided into two mainstream paradigms, including two-stage and one-stage detection frameworks. Inspired by R-CNN [35] and Fast R-CNN [36], Ren *et al.* [23] proposed Faster R-CNN, which laid down an original end-to-end two-stage processing framework. Also, the detection accuracy has been improved continually by numerous successors [25], [37], [38]. In contrast to improving detection accuracy regardless of time consumption, A considerable detection speed without much performance degradation is pursued by one-stage detectors [24], [39]–[41].

With respect to aircraft detection in SAR images by using CNN-based methods, Wang *et al.* [19] and Guo *et al.* [20] employed a CNN-based classifier to identify aircraft's candidates within suspicious areas. To regress locations of aircraft accurately, Diao *et al.* [21] employed a Fast R-CNN to discriminate aircraft. Considering partial characteristic of aircraft, He *et al.* [22] introduced a component-based detector based on YOLOv2 [26], which consists of paralleled root and part detectors, to identify and match fuselages and wings of aircraft. Besides, Guo *et al.* [42] proposed an attention pyramid network based on the feature pyramid network (FPN) to explore scattering information enhancement (SIE) of aircraft. To handle sparse and discrete scattering points of aircraft, Zhao *et al.* [43] introduced a pyramid attention dilated network (PADN) based on RetinaNet [44] by designing a multibranch dilated convolutional module (MBDCM). Also, an attention feature fusion network (AFFN) [45] is designed for aircraft detection in SAR images. Undoubtedly, aircraft could be detected by these methods stably benefiting from the powerful feature representation ability of CNN. However, the lacks of full considerations of aircraft's domain knowledge and the challenges limit further promotion boosted by these methods.

III. Methodology

In this section, key inspirations of our method for tackling the aforementioned problems are described in Section III-A. Also, the holistic architecture of our method and its inner modules, including AFFM, DLCM, and ADM, is depicted in Sections III-C, III-D, and III-E, respectively. Loss functions are provided at the end of this section.

A. Motivations

Feature refinement and alignment are the main concentrations of our method for tackling domain knowledge and challenges of aircraft detection in SAR images.

1) Feature Refinement: A balanced representation of aircraft's low-level textural details and high-level semantic features with powerful suppression of interference is essential for detecting aircraft effectively. Although the detailed information of small aircraft could be extracted by a low-level feature map, the semantic information it contains is rare and the correlations among different aircraft's features are weak. A wider receptive field could be obtained by a deeper neural network. However, textural details of aircraft may be indistinguishable due to a long propagation pathway, in which several intermediate convolutions involved. Instead of employing additional bottom-up pathways [46], [47], combining low-level textural details and high-level features directly could acquire multi-level aircraft's features distinctly without losing much original textural information. In addition, serious interference caused by complex surroundings should also be resolved. Although significant features could be refined and highlighted [48]–[50], a layer-aware attention mechanism should be specially cared considering the differences of aircraft's characteristics across multilevel feature channels.

2) Feature Alignment: A highly constant alignment between discrete backscattering features and convolutional spots is essential for locating and identifying aircraft precisely. In comparison to traditional convolution, of which convolutional kernels' sampling locations are fixed, e.g., 3×3, deformable convolution [51], [52] could extract features within irregular areas by using convolutional kernels, of which sampling locations are tuned by trainable offsets. Hence, it is naturally suitable for representing discrete deformable features of aircraft accurately with the alleviation of deformable convolution in our method. Besides, a progressive detector

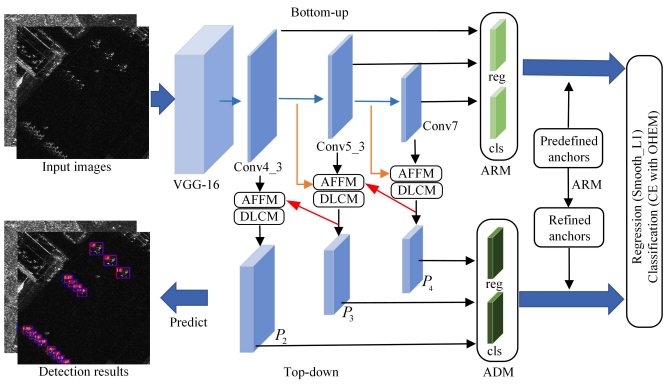

Fig. 3. Architecture of AFRAN.

striking the tradeoff between accuracy and speed is practical and essential for detecting sparsely parked small aircraft in SAR images.

Shortly, the low-level textural and high-level features of aircraft are fused and emphasized thoroughly by feature refinement. Also, the significant discrete features of aircraft are captured precisely by feature alignment. With the aforementioned two criteria as guidelines, our method could perform competitively.

B. Overall Architecture of AFRAN

Our method belongs to a one-stage detector, of which the architecture is shown in Fig. 3.

1) Bottom-Up Pathway: A truncated VGG-16 network [53] acted as a backbone is employed to extract fundamental features from the input SAR images in the bottom-up pathway. Three intermediate layers of VGG-16, including Conv4_3, Conv5_3, and Conv7, are selected as hierarchically semantic features of aircraft. The spatial sizes of the three feature maps are 8, 16, and 32 times downsampling of the original input images, which are 80×80, 40×40, and 20×20 in pixels, respectively.

2) Top-Down Pathway: A fine-grained feature pyramid is established by combining and refining the three-layer features selectively in the top-down pathway. Three groups of AFFM are first adopted to fuse and filter semantic information from the three-layer feature maps. Here, two feature forward propagation pathways (orange arrows in Fig. 3) are introduced to fully utilize low-level textural details of aircraft. Considering much interference around aircraft may be extracted by traditional convolution with axis-aligned kernels, three groups of DLCM are introduced following AFFM to capture aircraft's discrete characteristics adaptively. Most specifically, P_4 is set up at the first group of AFFM and DLCM by leveraging Conv_7 and Conv5_3. P_3 is constructed by the second group of AFFM and DLCM by using Conv5_3, Conv4_3, and P_4 together. Also, the low-level fine-grained feature map P_2 is established by fusing Conv4_3 and P_3 at the third group of AFFM and DLCM.

3) Coarse-to-Fine Detection Head: Identifying aircraft progressively could alleviate mismatches between initial anchors and ground truths on location and quantity. Drawing from a coarse-to-fine detection strategy [27], an anchor refinement module (ARM) and an improved object detection module, namely, ADM [54], are introduced to our method for predicting aircraft progressively. Specifically, refined anchors are first acquired by adjusting initial anchors based on parameterized locations and confidences, which are predicted by feeding Conv4_3, Conv5_3, and Conv7 to ARM. Aircraft are predicted eventually by feeding features, where locations at P_2, P_3, and P_4 are calculated by refining anchors, to ADM subsequently.

To optimize our method, a multitask loss, including binary and multiclass cross-entropy losses with online hard example mining (OHEM) [55] and smooth L1 loss, is employed for supervising classification and regression branches of ARM and ADM, simultaneously.

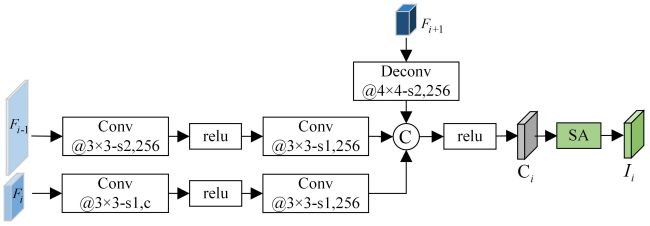

Fig. 4. Structure of AFFM.

C. Attention Feature Fusion Module

To leverage aircraft's low-level textural details and semantic features harmoniously, an AFFM shown in Fig. 4 is introduced when building the fine-grained feature pyramid of AFRAN.

1) Multilevel Feature Aggregation: To acquire multidimension features, a 3×3 convolution with stride of 2 followed by a relu activation function is first applied at the $i-1$th basic feature map F_{i-1} to reduce its spatial size for feature fusion. Meanwhile, the $i+1$th basic feature layer F_{i+1} is upsampled by a 4×4 deconvolution to acquire abundant and discriminative semantic features, which are helpful for detecting small aircraft from low-level feature layers. To satisfy requirements of different modules, one and two layers of 3×3 convolution layers with a stride of 1 are attached at the $(i-1)$th feature map F_{i-1} and the ith feature map F_i, respectively, to maintain discrepancy as well as relaxing interference of different feature stages. After that, the modified F_{i+1}, F_{i-1}, and F_i with the same spatial sizes are obtained. After concatenating (©) the three intermediate features along channel dimension, an intermediate feature map C_i containing abundant textural and semantic features of aircraft is obtained. Specifically, Conv5_3 and Conv7 are adopted to build the topmost intermediate feature C_4. The middle intermediate feature C_3 is constructed by leveraging Conv4_3, Conv5_3, and P_4. The low-level intermediate feature map C_2 is obtained by combining Conv4_3 with P_3 finally.

2) Feature Refinement by SA: Although abundant features of aircraft are accumulated after concatenation, gaps across different feature layers are still obvious. Also, interference of uninformative background around aircraft may also be collected. To highlight significant features of aircraft and suppress background interference at different feature layers

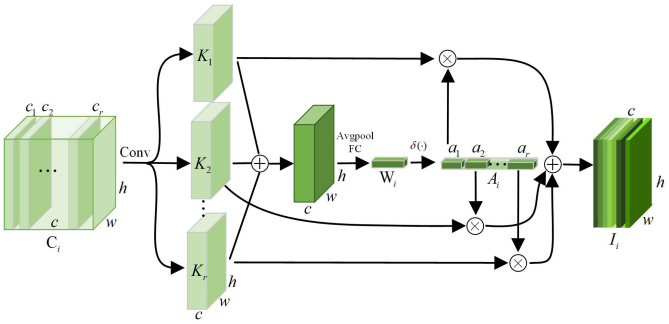

Fig. 5. Structure of SA.

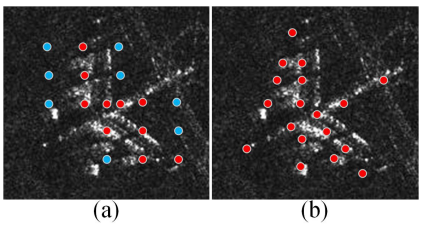

Fig. 6. Sampling locations of convolutional kernels for aircraft and surroundings by different types of convolution. (a) Locations sampled by vanilla convolutional kernels. (b) Locations sampled by deformable convolutional kernels.

carefully, a layer-aware attention mechanism, namely, SA [56], is employed following the concatenated feature C_i. The structure of SA is shown in Fig. 5.

Let $\mathbf{C}_i \in \mathbf{R}^{h\times w\times c}$ denote a concatenated feature block of r feature maps, where h, w, c, and r refer to block's height, width, channel number, and the number of concatenated features within AFFM, respectively, in our method. To acquire refined feature block I_i, r groups of individual feature maps $(\boldsymbol{K}_1, \boldsymbol{K}_2, \ldots, \boldsymbol{K}_r)$, of which numbers of channels all equal to c, are first generated by sending $\mathbf{C}_i$ to a convolution (Conv) with $c\times r$ filters and splitting along channel dimension. Then, these r groups of features are added elementwisely ($\oplus$) and sent into an adaptive two-dimensional average pooling (Avgpool) and fully connection (FC) in sequence to produce a weighted vector $\mathbf{W}_i$. It represents the contributions of different channels of the concatenated feature map. Then, r groups of channel attention weights a_1, a_2, …, a_r derived from $\mathbf{A}_i$ is acquired by applying a softmax function (σ) on $\mathbf{W}_i$. Finally, the layer-aware refined feature map $\mathbf{I}_i \in \mathbf{R}^{h\times w\times c}$ could be acquired by an elementwise addition ($\oplus$) of r groups of productions ($\otimes$) between K_r and the corresponding vector a_r.

D. Deformable Lateral Connection Module

Representing discrete yet constant features of aircraft by a suitable convolution is essential for accurate aircraft detection in SAR images. As a toy example shown in Fig. 6, red and blue points refer to sampling locations of convolutional kernels for aircraft and surroundings, respectively. As shown in Fig. 6(a), backscattering information of surroundings around aircraft is easily captured by a traditional convolutional operation due to its regular and rigid sampling strategies. However, as shown in Fig. 6(b), a deformable convolution [51], [52] is naturally appropriate for capturing discrete and significant features of aircraft without introducing much background interference benefiting from its deformable convolutional strategy. Based on the superiority, three groups of DLCM with several deformable convolution stacked, as shown in Fig. 7, are designed in our method for capturing aircraft's discrete features following AFFM.

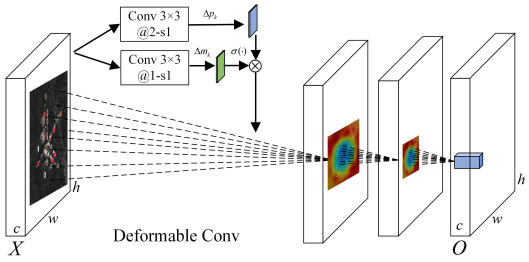

Fig. 7. Structure of DLCM.

As shown by the blue and green rectangles of Fig. 7, two vanilla 3×3 convolutions with stride 1 and outputs 2 and 1 are first attached on the input feature map X to generate a two-dimensional offset map Δp_k and a score mask Δm_k, respectively. Therefore, revised sampling locations of the deformable convolution could be acquired by adjusting original axis-aligned convolutional spots through Δp_k. Also, discrete deformable features of aircraft are generated after modulating deformable values with the score mask Δm_k. In our method, flexible and irregular information of aircraft could be represented effectively by stacking several deformable convolution in DLCM sequentially.

Concisely, the deformable convolution is formulated in (1). Here, $\mathbf{X}(p_n + p_k + \Delta p_k)$ refers to deformable input features. They are acquired by adjusting original eight-neighborhood features $\mathbf{X}(p_n + p_k)$ of $\mathbf{X}(p_n)$ using offsets Δp_k, where $k = (i, j)\ i, j \in -1, 0, 1$ for a 3×3 convolutional kernel. The final output matrix $\mathbf{Y}(p_n)$ is obtained after mapping and modulating $\mathbf{X}(p_n + p_k + \Delta p_k)$ through trained weights $\mathbf{W}_k$ and modulation factor Δm_k simultaneously

$$\mathbf{Y}(\mathbf{p_n}) = \sum_{\mathbf{k}}^{\mathbf{K}} \mathbf{W_k} \cdot \mathbf{X}(\mathbf{p_n} + \mathbf{p_k} + \Delta \mathbf{p_k}) \cdot \Delta m_k. \tag{1}$$

E. Anchor-Guided Detection Module

Due to the lack of well-designed feature cropping strategies, e.g., ROI Align [57] and ROI Wrapping [23] leveraged in two-stage detectors, one-to-many mappings and obvious misalignments between anchors and features in one-stage detectors easily incur inaccurate identification and location for small aircraft. Instead of representing aircraft by a set of representative points [58] or features aligned by initial anchors [59], an ADM derived from [54] is adopted in our method for establishing a strict single-mapping between refined anchors and features. Most specifically, unique features calculated by refined anchors are utilized for identifying aircraft ultimately.

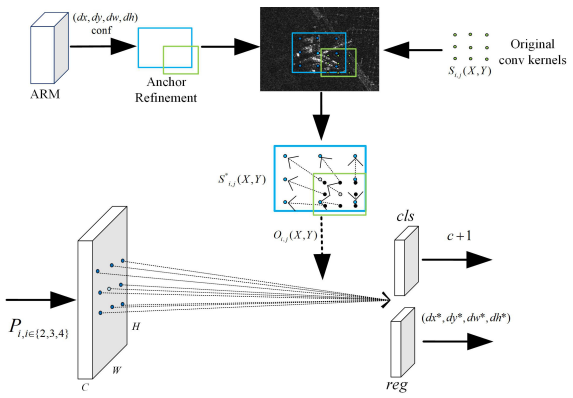

Fig. 8. Structure of ADM.

As shown in Fig. 8, blue rectangle refers to a refined anchor generated by adjusting locations and suppressing redundancy of initial anchors based on parameterized offsets (dx, dy, dw, and dh) and confidence produced by ARM. It is distinctly that features covered by a refined anchor are more representative than those covered by an initial anchor (the green rectangle in Fig. 8) resulting from an improved overlap between the revised anchor and the ground truth aircraft. Thus, a higher correspondence between features captured by the revised sampling points $S^*_{i,j}(X, Y)$ (nine blue points in Fig. 8) and the aircraft could be established than before. Based on advantages of deformable convolution, two-dimensional offsets $O_{i,j}(X, Y)$ between revised sampling points ($S^*_{i,j}(X, Y)$) and the related original sampling points ($S_{i,j}(X, Y)$) could be calculated and acted as deformable convolutional offsets for final regression and classification.

Concisely, the processing flows of ADM are formulated as (2), (3), and (4). The original sampling points $S_{i,j}(X, Y)$ are formulated as (2), where $i, j \in \{0, 1, k-1\}$ for a 3×3 convolution

$$S_{i,j}(X, Y) = \left(X - \left\lfloor \frac{k}{2} \right\rfloor + i + 0.5, Y - \left\lfloor \frac{k}{2} \right\rfloor + j + 0.5\right). \quad (2)$$

The corresponding aligned sampling points $S^*_{i,j}(X, Y)$ at feature maps $\mathbf{F}$ are formulated as (3), where $x1$, $y1$, $x2$, and $y2$ refer to the locations of refined anchors

$$S^*_{i,j}(X, Y) = \left(\frac{kx_1 + (x_2 - x_1)(i + 0.5)}{kS}, \frac{ky_1 + (y_2 - y_1)(j + 0.5)}{kS}\right) \quad i, j \in 0, 1, 2, 3 \ldots k-1. \quad (3)$$

The two-dimensional offsets for a deformable convolution formulated as (4) are obtained by subtracting (3) from (2)

$$\begin{aligned} O_i(X) &= S^*_{i,j}(X) - S_{i,j}(X) \\ &= \frac{x_1}{S} - X + \left\lfloor \frac{k}{2} \right\rfloor + \left(\frac{x_2 - x_1}{S} - 1\right)(i + 0.5) \\ O_j(Y) &= S^*_{i,j}(Y) - S_{i,j}(Y) \\ &= \frac{y_1}{S} - Y + \left\lfloor \frac{k}{2} \right\rfloor + \left(\frac{y_2 - y_1}{wS} - 1\right)(j + 0.5). \end{aligned} \quad (4)$$

F. Loss Functions

The overall loss for optimizing our network provided by (5) contains losses of ARM and ADM. As defined by (6), each part of (5) is a weighted sum of the classification loss (L_{conf}) between predicted classes (x) and truth labels (c) and the regression loss (L_{reg}) among predicted classes (x), bounding boxes (l), and ground truth locations (g)

$$L_{\text{total}} = L_{\text{ARM}} + L_{\text{ADM}} \quad (5)$$

$$L(x, c, l, g) = \frac{1}{N}\left(L_{\text{conf}}(x, c) + \alpha L_{\text{reg}}(x, l, g)\right) \quad (6)$$

where N refers to the number of positive anchors matched to any ground truth objects. As defined by (7), L_{conf} is a binary or multiclass cross-entropy loss between the predicted classes x and ground truth labels c

$$L_{\text{conf}}(x, c) = -\sum_{i \in \text{Pos}}^{N} x^p_{i,j} \log\left(\hat{c}^p_i\right). \quad (7)$$

The Iverson bracket indicator function $x^p_{i,j}$ is 1 where the ith anchor matches to jth ground truth object for class p and otherwise 0. $\hat{c}^p_i$ refers to a softmax loss defined as follows:

$$\hat{c}^p_i = \frac{\exp\left(c^p_i\right)}{\sum_p \exp\left(c^p_i\right)}. \quad (8)$$

For regression, L_{reg} formulated as (9) refers to a smooth L1 loss of offsets between predicted bounding boxes (l) and ground truths (g), where c_x, c_y, w, and h are centers, widths, and heights of initial or refined anchors and matched ground truths

$$L_{\text{reg}}(x, l, g) = \sum_{i \in \text{Pos}}^{N} \sum_{m \in \{c_x, c_y, w, h\}} x^k_{ij} \, \text{Smooth}_{\text{L1}}\left(l^m_i - \hat{g}^m_j\right). \quad (9)$$

In the training phase, the initial anchors are first revised by predictions of ARM. Only refined anchors with higher confidence than a preset threshold θ will contribute loss of ADM.

IV. Experiments and Analysis

A. Datasets and Experimental Setup

1) Datasets Descriptions: Since there is no publicly available dataset for aircraft detection in SAR images, a self-built aircraft sliced dataset and a large scene SAR image are collected for investigating the detection performance of our method in the experiments.

1) *Aircraft Sliced Dataset:* A self-built aircraft-sliced dataset is constructed by using 174 large scene SAR images collected from Chinese GF-3 and German TerraSAR-X satellites. The large scene images captured by GF-3 satellite working in C-band HH and VV polarization and SpotLight (SL) observing mode are with a nominal resolution of 1.0 m. Another images captured by TerraSAR-X satellite working in X-band and SL mode are with a nominal resolution of 0.5 m. The ground truth aircraft were manually annotated by experts of SAR ATR by considering both prior knowledge and the

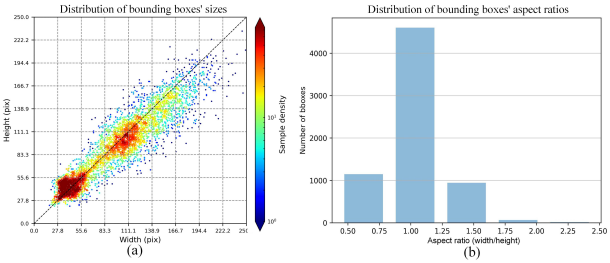

Fig. 9. Distributions of the bounding boxes' and aspect ratios of the self-built aircraft sliced dataset. (a) Distribution of bounding boxes' sizes. (b) Distribution of bounding boxes' aspect ratios.

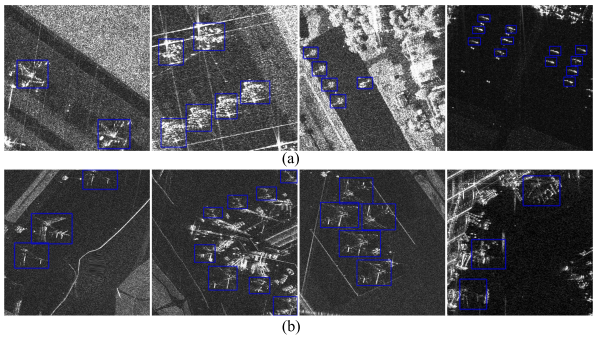

Fig. 10. Slices of SAR images in the self-built aircraft sliced dataset. (a) Aircraft imaged by GF-3 satellite. (b) Aircraft imaged by TerraSAR-X satellite.

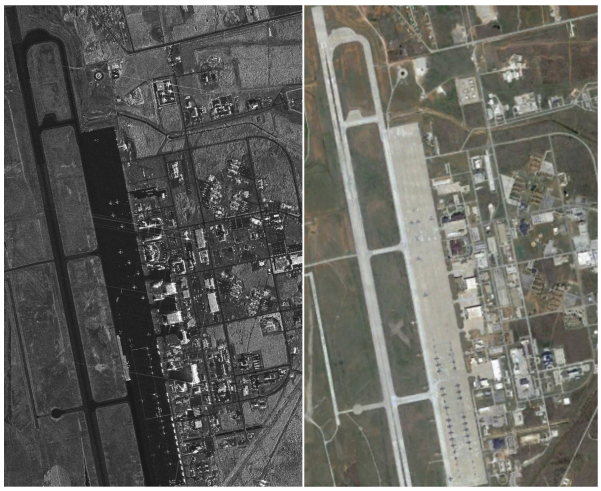

Fig. 11. Large scene for aircraft detection.

corresponding optical images. After random cropping, 2317 nonoverlapped 640×640 slices are collected with 6781 aircraft, of which structures, outlines, and main components are clear and wings ranging from about 15 to 75 m in total. The distributions of bounding boxes' sizes in pixels and aspect ratios are given by Fig. 9(a) and (b), respectively. Also, some slices of our dataset are shown in Fig. 10(a) and (b), in which real aircraft encompassed by blue rectangles are imaged by GF-3 and TerraSAR-X satellites, respectively.

2) *Large Scene SAR Image:* A large scene image is also adopted for evaluating the performance of our method. The left and the right subfigures of Fig. 11 illustrate the large scene image, including an airport and corresponding optical image cropped from Google Earth software, respectively. Some imaging parameters are listed concisely in Table I.

TABLE I
IMAGING PARAMETERS OF THE LARGE SCENE IMAGE

Parameter name	Value
Product level	Level 2
Product serial number	3253042
Product resolution	1.0 (meter)
Imaging central location	(99.82525 W, 32.412828 E)
Product height and width	25784×23161 (pixel)

2) Implementation Details: The training, valid, and test sets are constructed by dividing the original slices with a ratio of 5:2:3. Thus, numbers of images in the three sets are 1158, 463, and 696, respectively. Data augmentation strategies, including contrast, illumination distortion, mirroring, random flipping, expanding, and cropping, are adopted to increase diversity of aircraft in the training set. To maintain considerable overlaps between initial anchors and aircraft, the basic anchor scales for P_2, P_3, and P_4 of our method are set to 32, 64, and 128, respectively. Also, three aspect ratios $\{0.5, 1.0, 2.0\}$ are assigned to each anchor for all pyramid levels. Our method is trained for 200 epochs with a minibatch of 4. The initial learning rate is $1e^{-3}$ and decayed at 75 and 150 epochs with rate (γ) 0.1. To achieve a stable convergence, warm-up is enabled at the first five epochs. Stochastic gradient descent (SGD) with weight decay rate $5e^{-4}$ and momentum 0.9 is adopted to optimize network parameters. Other compared methods are implemented based on the mmdetection framework [60].

3) Evaluation Metrics: In all experiments, six average precision indicators from Microsoft COCO [13] are adopted, including AP, $AP^{.5}$, $AP^{.75}$, AP^{s}, AP^{m}, and AP^{l}, to judge performance of different methods. Moreover, $AP^{.5}$ and $AP^{.75}$ evaluate the AP scores at 0.5 and 0.75 intersection of union (IoU) thresholds between predictions and ground truths, respectively. AP^{s}, AP^{m}, and AP^{l} refer to AP scores of methods for detecting small, middle, and large aircraft averaged by ten IoU thresholds (0.5, 0.55, . . . , 0.95). In addition, precision (P), recall (R), and F score (F_1) are also adopted. The six AP metrics are calculated at 0.05 confidence threshold. Precision, recall, and F_1 are acquired at 0.5 confidence threshold. All these metrics are acquired at 0.5 IoU threshold. In terms of time and space complexities, frame-per-second (FPS), model parameter volumes (Params), and multiply-accumulate operations (MACs) are employed, of which the last two metrics are defined as follows:

$$\text{Params} = C_{\text{out}} \cdot (k_w \cdot k_h \cdot C_{\text{in}} + 1). \tag{10}$$

C_{out}, C_{in}, k_w, and k_h in (10) are the output, input channels, and kernel sizes of a convolution

$$\text{MAC} = C_{\text{out}} \cdot C_0 \cdot k_w \cdot k_h \cdot H_o \cdot W_o. \tag{11}$$

Here, k_w, k_h, C_{out}, and C_{in} are width, height, output, and input channels of a convolution. H_o and W_o are spatial sizes of an output feature map.

TABLE II
EFFECTS OF INNER MODULES

Models	FF	SA	DLCM	ADM	P	R	F_1	AP	$AP^{.5}$	$AP^{.75}$	AP^s	AP^m	AP^l	FPS
RefineDet	-	-	-	-	0.815	0.935	0.871	0.530	0.932	0.547	0.388	0.521	0.549	60
Baseline*	-	-	-	-	0.829	0.933	0.878	0.520	0.933	0.535	0.385	0.515	0.536	**63**
	✓	-	-	-	0.827	0.926	0.874	0.522	0.932	0.517	0.371	0.508	0.543	58
	✓	✓	-	-	0.871	0.931	0.900	0.536	0.941	0.563	0.427	0.524	0.555	50
	✓	✓	✓	-	0.899	0.927	0.913	0.538	0.929	0.572	0.394	0.526	0.564	46
AFRAN (Ours)	✓	✓	✓	✓	**0.904**	**0.932**	**0.918**	**0.554**	**0.941**	**0.597**	**0.481**	**0.537**	**0.576**	40

* The baseline detector is a modified AFRAN, which only contains three layers of multi-scale features for prediction. By enabling different sub-modules progressively, performance for detecting aircraft in SAR images is improved. Bold texts indicates the best value in each row.

In addition, precision–recall curves of different AP metrics and visual detection results are also provided for further judgments.

B. Ablation Studies

1) Quantitative Analysis: The quantitative contributions of different inner modules are evaluated by enabling them progressively and the results are shown in Table II. A baseline detector is constructed by removing the three modules from AFRAN but maintaining the basic architecture, i.e., the three-layer fine-grained feature pyramid. Besides, we disentangle AFFM into feature fusion especially feature forward (orange arrows in Fig. 3) and SA block, donated as FF and SA, and explore their contributions to aircraft's textural feature extraction and significant feature refinement in detail. Considering the similar architecture and processing scheme between RefineDet and our method, the detection results of RefineDet are also provided in the first column of Table II. According to Table II, several insightful findings could be summarized as follows.

2) Baseline: The high-level semantic features play an important role for identifying and locating large aircraft. In comparison to the detection results of RefineDet, the precision rate of the baseline increases to 0.829 and is 1.4% higher than that of RefineDet at 0.5 IoU threshold. It may be because less background information is accumulated at low-level feature layers through the top-down pathway, which reduces interference for aircraft detection to some extent. However, aircraft's high-level semantic features extracted by the baseline are restricted resulting from the three-layer feature pyramid constructed by a truncated VGG-16 of the baseline, which weakens its ability for identifying aircraft accurately. Therefore, distinct decreases emerge on AP, $AP^{.75}$, and AP^l when detecting aircraft by baseline, which are 1.0%, 1.2%, and 1.3% lower than those of RefineDet.

3) Effects of FF: By merely propagating aircraft's textural details forwardly, uninformative background interference may also be accumulated further, which leads to much confusion for aircraft discrimination. Specifically, the score of $AP^{.75}$ achieved by baseline (w/FF) (the third column of Table II) decreases obviously and is 1.5% lower than that of baseline.

4) Effects of SA: Aircraft's low-level textural details could be utilized effectively benefiting from powerful feature refinement ability of SA blocks, which promotes the detection performance of our method greatly. In comparison to the detection results of baseline (w/FF) and baseline (w/FF+SA) (the third and the fourth columns of Table II), scores of precision, F_1, AP, $AP^{.75}$, AP^s, AP^m, and AP^l achieved by baseline (w/FF+SA) are 4.4%, 1.6%, 1.4%, 4.6%, 5.6% 1.6%, and 1.2% higher than those of baseline (w/FF). It fully verifies the effectiveness of SA blocks for refining aircraft's significant features and suppressing uninformative background. To inspect the effects of SA blocks further, values of different fine-grained feature channels w/o or w/SA blocks are also provided in Fig. 12 in blue and red curves, respectively. Meanwhile, the top-k (k=20) corresponding feature channels sorted by activation values are also marked out with blue and red real dots. Obviously, peaks of feature channels after SA blocks (red points) distribute sparsely, however, densely and locally for peaks of initial feature channels, which illustrates that a wider range of aircraft's significant features across multilevel feature maps could be boosted. Especially for building the middle-level fine-grained feature map, the first 256 channel indexes at Fig. 12(b) are enhanced and a more balanced feature map could be obtained after enabling SA block, which proves powerful feature refinement ability of SA block.

5) Effects of DLCM: Feature alignment by DLCM at lateral connections of the fine-grained feature pyramid promotes our method's ability for locating aircraft especially small aircraft further. Benefiting from powerful deformable modeling ability of DLCM, aircraft's high-level discrete features could be perceived accurately, leading to improvements on precision (P), $AP^{.75}$ and AP^l. However, scores of AP^s encounter a sharp decrease and are 3.3% lower than those of baseline (w/FF+SA). It might be because of insufficient semantic information and complex background interference exists at low-level feature maps, which restricts DLCM's ability for capturing aircraft's discrete and significant information.

6) Effects of ADM: A tight correspondence between refined anchors and features is constructed by baseline (w/FF+SA+DLCM+ADM), which significantly boosts the performance of our method for detecting aircraft. Compared with detection results of baseline (w/FF+SA+DLCM) (the fifth row of Table II), of which the detection head is the same as that of RefineDet, remarkable improvements on all indicators are acquired by baseline (w/FF+SA+DLCM+ADM) (the last column of Table II). Specifically, scores of AP, $AP^{.75}$, and AP^s are 1.6%, 2.5%, and 8.7% superior to those of baseline (w/FF+SA+DLCM), respectively.

Furthermore, the number of positive anchors and the values of average and maximal IoU between initial anchors and

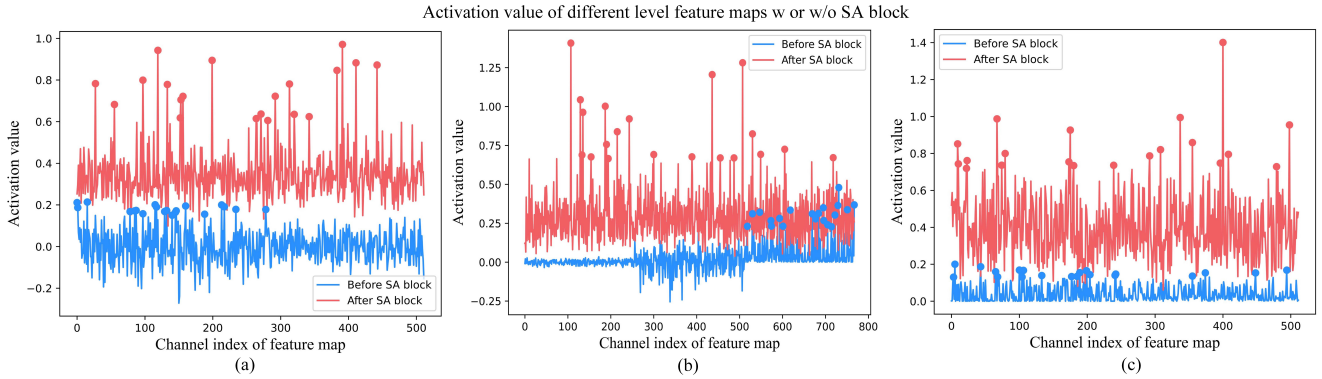

Fig. 12. Activation values of different levels of fine-grained feature maps w/ or w/o SA blocks. Activation values of low-, middle-, and high-level fine-grained feature maps are plotted in (a), (b), and (c), respectively.

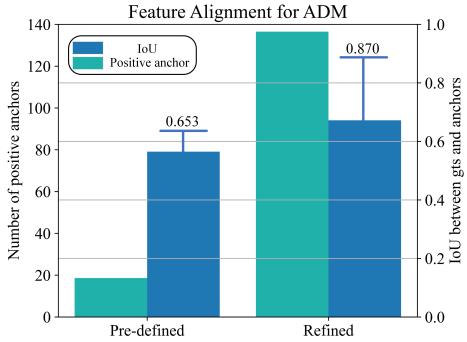

Fig. 13. Feature alignment of refined anchors for ADM. Blue and green bars show average IoU scores and positive anchor numbers achieved by initial and refined anchors at 0.5 IoU threshold, respectively. The horizontal lines indicate the maximal IoU scores acquired by initial and refined anchors.

ground truths or refined anchors and ground truths are given in Fig. 13. Obviously, more positive anchors with higher IoU score are acquired by refined anchors than those of initial anchors. The average maximal IoU score between ground truths and refined anchors is 0.870 yet 0.653 for initial anchors, which further indicates positive guidance of refined anchors for ADM.

7) Qualitative Analysis: To further investigate the evolution of features maps acquired by the baseline equipped with the three submodules progressively, aircraft detection results and heat maps of specific fine-grained feature maps, including P_3 and P_4, are shown in Fig. 14. Some observations could be summed up as follows.

8) Clear Boundaries Between Aircraft and Background Are Obtained by Baseline (w/FF): Specifically, activation boundaries of small aircraft and surroundings at P_3 of baseline (w/FF) (the third column of Fig. 14) are more distinct than those of the baseline (the second column of Fig. 14). Also, aircraft's activation areas are converged at P_3 of baseline (w/FF) yet divergent at identical feature areas generated by RefineDet (the first column of Fig. 14). It might be because fine-grained features could be enhanced at P_3 further by fusing aircraft's local textural details from P_2. Besides, less uninformative background information is propagated from topper feature maps, i.e., P_4 in our method, which improves our method's ability for identifying aircraft and surroundings clearly.

9) Significant Components of Aircraft Are Specifically Highlighted by Baseline (w/FF+SA): According to the fourth column of Fig. 14, activation areas of small aircraft (dark blue areas encompassed by an orange dotted rectangle) are distinct and the background is homogeneous (areas in green color) by enabling SA blocks. It might be because of the effective layer-aware feature refinement ability provided by SA blocks that consistency of background and specificity of aircraft at different feature maps could be maintained when fusing and refining low-level textural details and high-level semantic characteristics of aircraft.

10) Clear Separations Among Densely Arranged Aircraft Are Obtained by Baseline (w/FF+SA+DLCM): According to the fifth column of Fig. 14, separations among densely parked small aircraft are clearer than those at the same feature map P_3 of baseline (w/FF+SA). Besides, activation areas of large aircraft at P_4 are more centered than those extracted by baseline (w/FF+SA), which promotes our model to locate aircraft precisely (also proved in Table II). It might be because DLCM could capture irregular information of aircraft adaptively and avoid extracting much uninformative background interference by employing deformable convolutions instead of traditional convolutions equipped with axis-aligned kernels.

11) Hierarchy and Consistency of Aircraft's Features at Multiscale Feature Maps Are Promoted by Baseline (w/FF+SA+DLCM+ADM): As shown by the last column of Fig. 14, activation areas of small aircraft are more consistent than those at the same feature maps of baseline (w/FF+SA+DLCM). Also, aircraft with different scales could be represented hierarchically by different feature levels, which decreases aliasing across different feature maps. Specifically, activation areas of small and large aircraft are represented distinctly at P_3 and P_4, respectively.

C. Comparing With CNN-Based Methods

1) Quantitative Analysis: The performance of our method is also evaluated by comparing with other CNN-based detectors on the test set of the self-built aircraft sliced dataset and a large scene image. Moreover, two methods specially designed

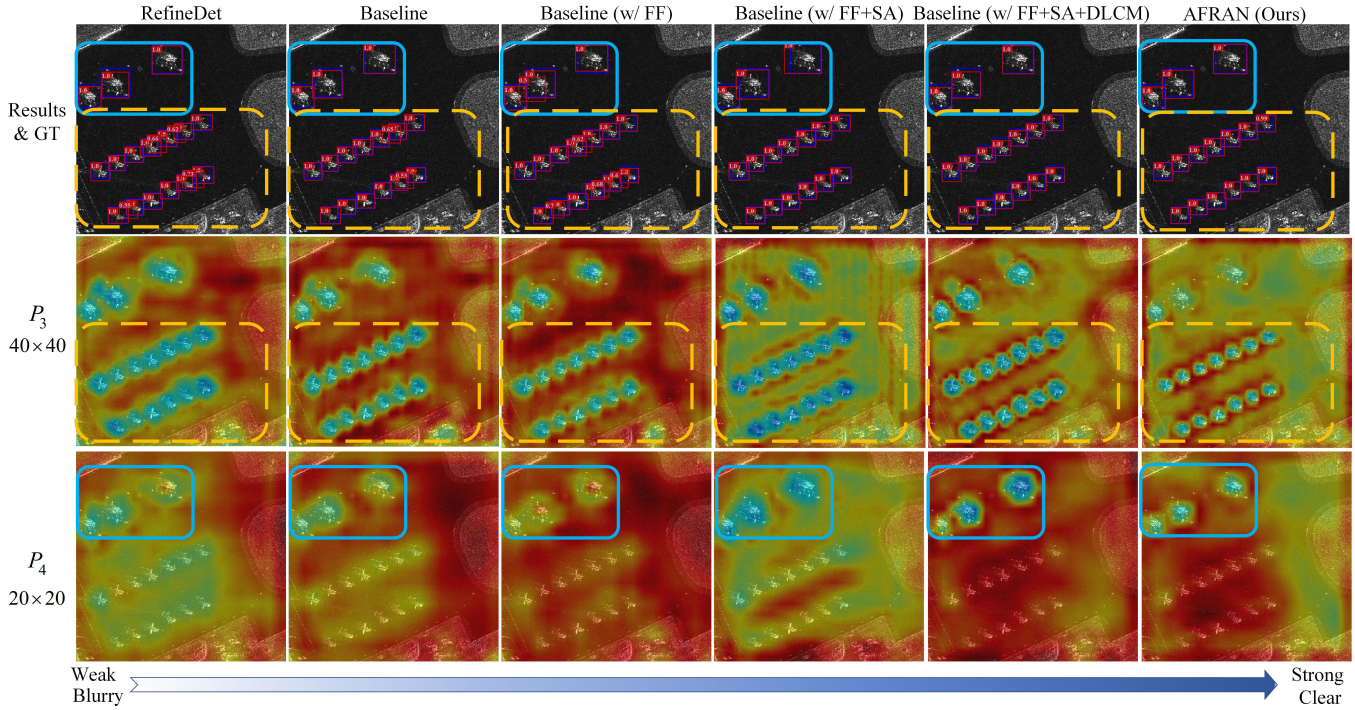

Fig. 14. Detection results and activation maps of specific fine-grained feature maps produced by our method equipped with the proposed submodules progressively.

TABLE III

DETECTION RESULTS OF DIFFERENT CNN-BASED METHODS ON TEST SET OF THE AIRCRAFT SLICED DATASET

Models	P	R	F_1	AP	$AP^{.5}$	$AP^{.75}$	AP^s	AP^m	AP^l
Two-stage methods									
Faster R-CNN [23]	0.893	0.890	0.892	0.520	0.877	0.567	0.331	0.495	0.561
FPN [25]	0.870	0.909	0.889	0.530	0.886	0.598	0.366	0.509	0.562
Cascade R-CNN [37]	0.898	0.889	0.893	0.539	0.876	0.594	0.398	0.517	0.577
DAPN [61]	0.885	0.830	0.857	0.448	0.851	0.413	0.250	0.420	0.489
One-stage methods									
SSD [24]	0.889	0.889	0.889	0.532	0.932	0.562	0.371	0.497	0.575
PADN [10]	0.842	0.916	0.878	0.512	0.906	0.518	0.308	0.480	0.554
RefineDet [27]	0.815	**0.935**	0.871	0.530	0.932	0.547	0.388	0.521	0.549
RPDet [58]	0.867	0.917	0.891	0.543	0.886	**0.612**	0.415	0.517	**0.584**
Ours (AFRAN)	**0.904**	0.932	**0.918**	**0.554**	**0.941**	0.597	**0.481**	**0.537**	0.576

for object detection in SAR images, including PADN [10] and dense attention pyramid network (DAPN) [61], are also reimplemented and compared together.

a) Detection accuracy: The detection accuracy of different methods on the test set of the self-built aircraft sliced dataset is shown in Table III. Specifically, AP and F_1 achieved by our method are 1.1% and 2.5% higher than those of the second-place methods (RepPoints Detector (RPDet) and Cascade R-CNN), respectively. Although a slight superiority on $AP^{.75}$ and AP^l is achieved by RPDet benefiting from its point-based aircraft representation strategy, however, its ability for identifying small aircraft is still unsatisfied. Besides, $AP^{.5}$ and AP^s scores achieved by RPDet are 5.5% and 6.6% lower than those achieved by our method, which are 0.941 and 0.481, respectively. It might be because of insufficient features of small aircraft extracted by the class-agnostic backbone. Obviously, competitive performance on metrics for evaluating location accuracy is achieved by most of the two-stage detectors benefiting from their advanced feature alignment strategies in comparison with classical one-stage detectors, e.g., SSD and RefineDet. However, our method could still locate aircraft competitively benefiting from its powerful feature alignment achieved by DLCM and ADM. In addition, an obvious promotion is obtained by our method comparing with the two domain-specific methods, including DAPN and PADN at all indicators. It might be because of their suboptimal architectures and limited utilization for aircraft's low-level details, which incurs a sharp degeneration for aircraft detection in SAR images.

b) Time and spatial complexities: The detection speed and spatial complexities are evaluated and given in Table IV. Clearly, benefiting from truncated VGG-16 backbone as well as balanced three-layer feature pyramid, our method only takes up 35.82M and 150.59G on parameter volumes (Params) and MACs, which is lower than those required by numerous two-stage detectors (Faster R-CNN, FPN, Cascade R-CNN, and DAPN) and some one-stage detectors (RPDet and PADN). Meanwhile, our method achieves higher detection speed than

TABLE IV
RUNNING TIME AND MODEL COMPLEXITY OF DIFFERENT CNN-BASED METHODS

Models	Backbone	Input size	FPS	Params(M)	MAC(G)
Two-stage methods					
Faster R-CNN [23]	ResNet-101	$\sim 1333 \times 800$	6	51.75	921.07
FPN [25]			16	60.04	290.35
Cascade R-CNN [37]			15	60.4	296.42
DAPN [61]			12	128.23	335.46
One-stage methods					
SSD [24]	VGG-16	$\sim 512 \times 512$	**90**	**25.22**	**88.84**
PADN [10]	ResNet-101	$\sim 640 \times 640$	33	102.16	247.12
RefineDet [27]	VGG-16	$\sim 640 \times 640$	63	34.05	150.00
RPDet [58]	ResNet-101	$\sim 1333 \times 800$	17	55.59	278.44
Ours (AFRAN)	VGG-16	$\sim 640 \times 640$	45	35.82	150.59

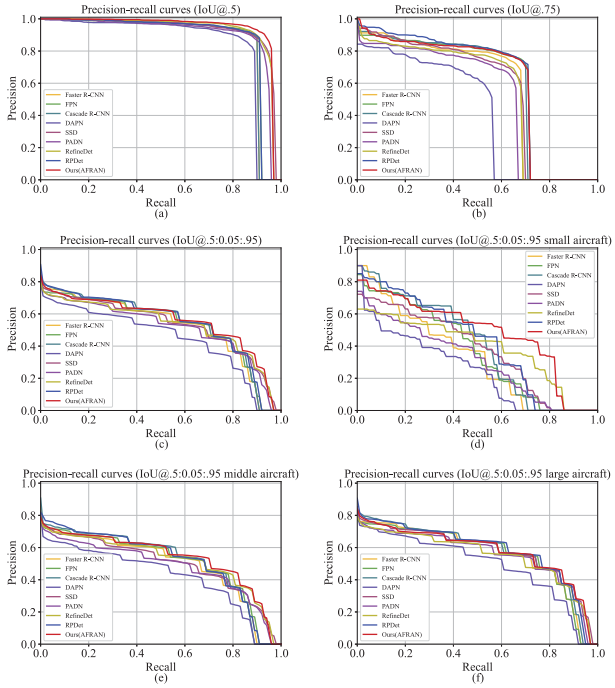

Fig. 15. Precision–Recall (PR) curves of detection results predicted by different CNN-based methods for detecting aircraft with different sizes at various IoU thresholds. (a) IoU at 0.5. (b) IoU at 0.75. (c) IoU from 0.5 to 0.95. (d) Small aircraft detection. (e) Middle aircraft detection. (f) Large aircraft detection.

RPDet and all two-stage detectors, which demonstrates its feasibility for real-time applications. However, our method runs slower than SSD and RefineDet causing by numerous parameters introduced by its complex submodules.

2) Qualitative Analysis: To inspect trends of detection performance, precision–recall curves of all compared CNN-based methods under various conditions of IoU thresholds and aircraft's sizes are shown in Fig. 15. Undoubtedly, the curve of our method [red curves in Fig. 15(a)–(f)] decreases much slightly than those of other methods with the increase of recall rate. Specifically, distinct advantages of our method exist at the conditions of 0.5 and 0.5 at 0.05:0.95 IoU thresholds, i.e., Fig. 15(a) and (b). Besides, the performance of our method is superior to other methods by a large margin for detecting small aircraft according to Fig. 15(d). It might be because of the full exploration and refinement of aircraft's multilevel features in AFFM, which further illustrates the powerful discrimination ability for aircraft in SAR images achieved by our method.

In addition, we further inspect the visual detection performance of different CNN-based methods at three representative situations and the results are shown in Fig. 16. In the first scene, aircraft with indistinct backscattering information are densely parked together. In comparison with obvious false alarms detected by RefineDet and RPDet and false negatives (the aircraft on the right side of the scene) missed by all methods except for PADN, our method detects these aircraft accurately with distinct separations. In the second scene, buildings on the left side of scene are detected as an aircraft by PADN, RefineDet, and RPDet. Also, aircraft with poor backscattering intensities (the first aircraft near the buildings) are missed by most compared methods. However, no false alarms are detected and false negatives are missed by our method. In the third scene, strong interference exists due to complex and angular lounge bridges as well as other ground facilities. In comparison to partial aircraft encompassed by other methods, e.g., Cascade R-CNN, PADN, and RefineDet, our method and RPDet all detect and encompass aircraft integrally benefiting from their flexible representation strategies for aircraft's discrete characteristics.

Apart from the above comparisons, the detection performance of different CNN-based methods over a large scene is investigated, as shown in Fig. 17. To evaluate the detection performance of different methods purely, only sliced cropping is adopted to satisfy input sizes of methods in advance. Also, the final detection results are acquired by mapping aircraft's coordination from partial slices to the large scene. Obviously, our method could detect all aircraft with fewer false alarms than other compared methods. Most specifically, due to the lack of efficient feature refinement operations after feature aggregation in FPN, Cascade R-CNN, DAPN, and PADN, surroundings within area A (the blue rectangle in Fig. 17) are easily recognized as aircraft. However, our method could discriminate aircraft and interference accurately. In terms of detecting aircraft with highly discrete backscattering information, e.g., aircraft parked at area B marked by the green rectangle in Fig. 17, aircraft's partial bodies are recognized as new ones by SSD, RefineDet, and RPDet. Also, ground

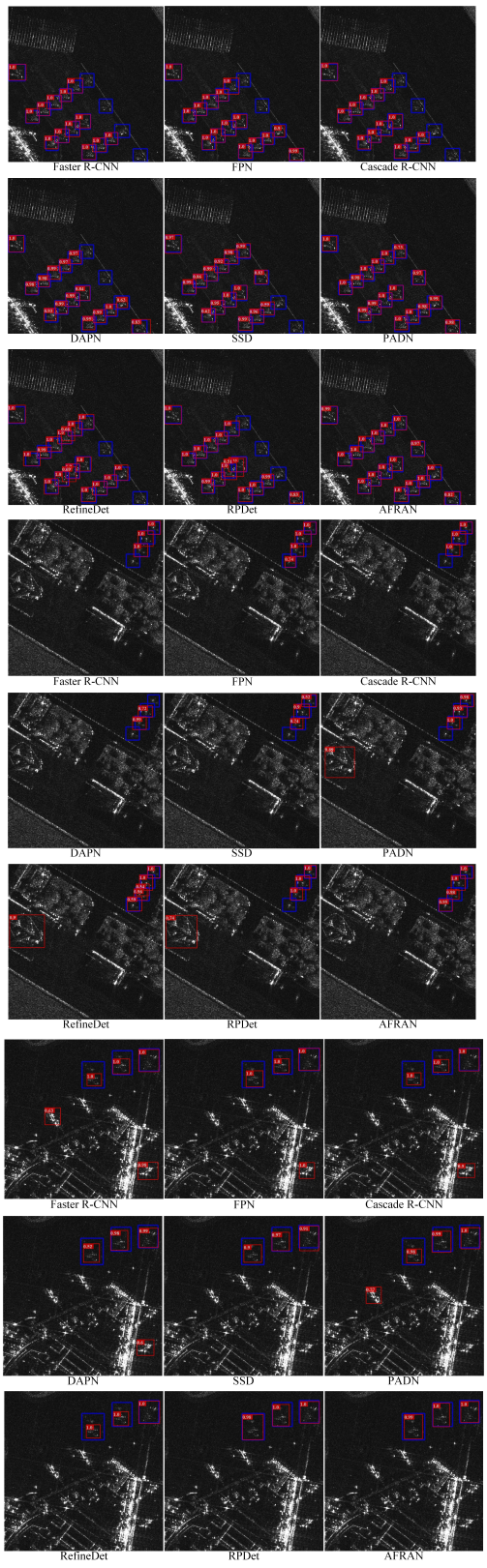

Fig. 16. Aircraft detection results of different CNN-based methods at three different situations.

facilities within airport are detected as aircraft by FPN, DAPN, and PADN and, however, discriminated accurately by our method without any false alarms and negatives.

D. More Discussion

Diversified parameter configurations also affect the performance of our method, which are discussed in this section.

1) Feature Forward of AFFM: In our method, two feature forward propagation pathways plotted as orange arrows in Fig. 3, are specially designed for enhancing representation ability of the fine-grained middle P_3 and the top P_4 feature maps for aircraft's low-level details. Table V shows the detection results of our method configured with different feature forward pathways in AFFM. For a fair reference, the detection results of AFRAN without any feature forward branches in AFFM acting as a baseline are given in the first row of Table V. Obviously, a 2.2% promotion on AP^s is achieved after combining low-level details of aircraft at Conv4_3 with semantic features existing at the middle layer Conv5_3 based on the baseline. However, a sharp decrease emerges on all indicators when merely introducing middle-level features to the topmost feature layer Conv7 due to the limited supervision of low-level details. When equipped with the two feature forward branches, a competitive performance, e.g., $AP^{.5}$ is 4.1% higher than that of the baseline, is acquired by our method benefiting from abundantly detailed and semantic features represented by the multiscale fine-grained feature pyramid.

2) Locations and Groups of SA Blocks: Due to various and complex semantic information at multiscale feature maps, how the SA blocks placed and the number of SA blocks are also worth exploration. Table VI shows the contributions of SA blocks located at different feature layers. Most specifically, enabling SA blocks at a single feature layer, i.e., the first four rows of Table VI, could boost the detection performance of our method, e.g., AP and F_1, resulting from powerful feature refinement donated by their well-designed layer-aware attentions. Especially, an obvious improvement is achieved after enabling SA blocks at the middle feature map when building P_3. It may be because of full consideration of aircraft's low- and high-level features remaining at the three feature maps. Besides, as can be seen from the fifth to the seventh columns of Table VI, most AP metrics increase when enabling SA blocks at two feature layers benefiting from effective utilization of complementary semantic features at different feature layers. Eventually, by applying SA blocks at all feature levels, AP, $AP^{0.75}$, and AP^s achieved by our method are 1.4%, 1.8%, and 7.6% higher than those of the baseline, which proves superior characteristics of SA blocks for feature refinement. In addition, the effects of the number of attention groups are shown at the eighth and ninth rows of Table VI. Due to an unspecific attention mechanism provided by the unified weights when setting attention groups of SA blocks to 1, a slight decrease on AP appears leashing their ability for refining semantic information at specific feature levels.

3) Receptive Fields of DLCM: Capturing discrete features of aircraft accurately calls for a sophisticated design of convolutional receptive fields. Table VII shows the detection results of our method configured with various dilation rates in DLCM. The larger the dilation rates are, the wider the receptive fields captured by DLCM. It could be observed that most of specific and comprehensive metrics, e.g., precision (P), AP^s, AP^m,

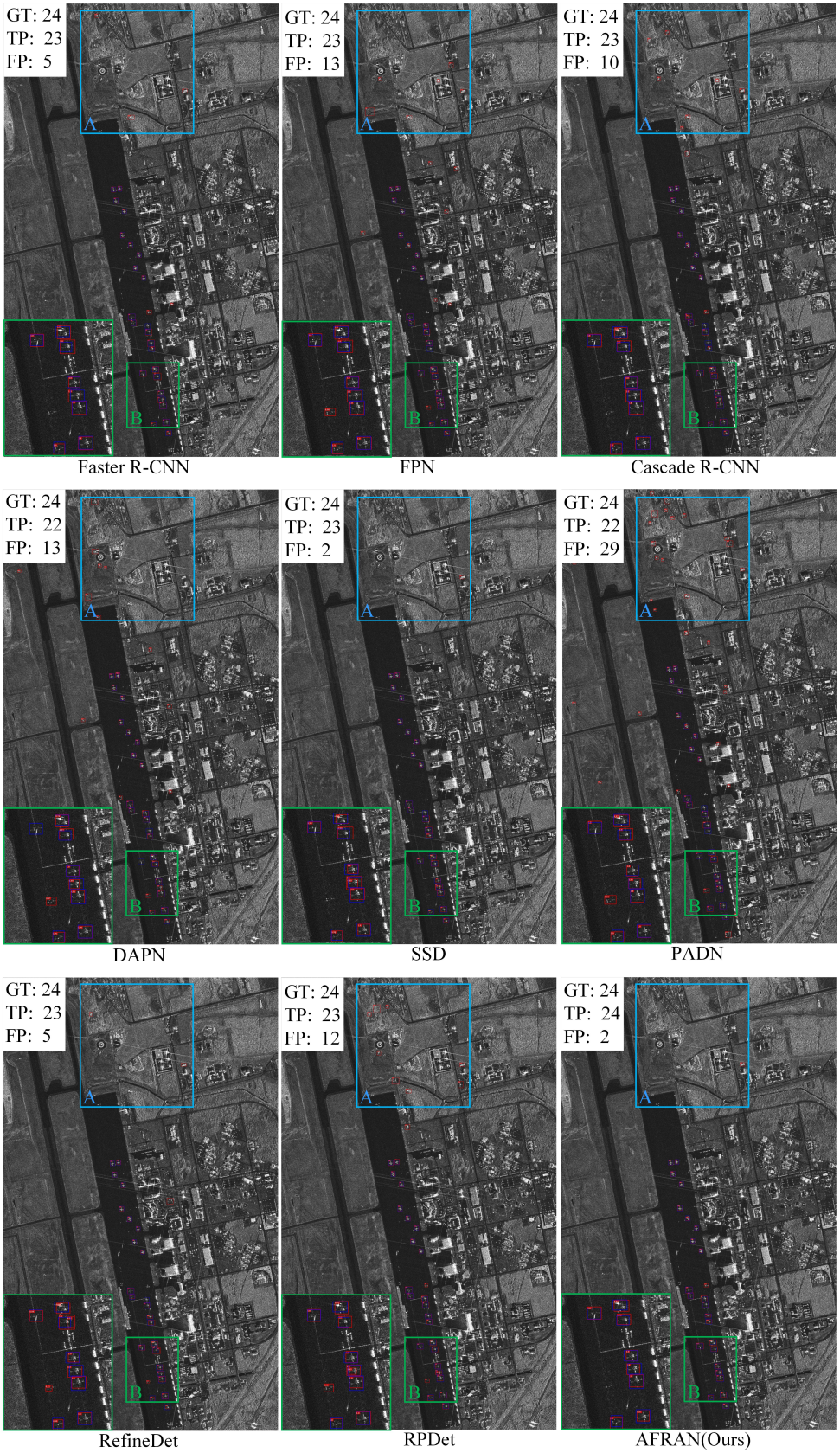

Fig. 17. Aircraft detection results over a large scene by different CNN-based methods.

TABLE V
FEATURE FORWARD CONFIGURATION OF AFFM

B → M	M → T	P	R	F_1	AP	$AP^{.5}$	$AP^{.75}$	AP^s	AP^m	AP^l
✗	✗	0.900	0.923	0.911	0.548	0.930	**0.607**	0.440	0.533	0.573
✓	✗	0.902	0.931	0.917	0.547	0.939	0.602	0.462	**0.538**	0.561
✗	✓	**0.963**	0.762	0.851	0.499	0.856	0.532	0.380	0.488	0.519
✓	✓	0.904	**0.932**	**0.918**	**0.554**	**0.941**	0.597	**0.481**	0.537	**0.576**

TABLE VI
LOCATIONS AND GROUP CONFIGURATIONS OF SA BLOCKS

B	M	T	P	R	F_1	AP	$AP^{.5}$	$AP^{.75}$	AP^s	AP^m	AP^l
-	-	-	0.883	0.932	0.907	0.540	0.939	0.579	0.405	0.520	0.568
✓	-	-	0.887	0.924	0.905	0.544	0.928	0.589	0.441	0.534	0.561
-	✓	-	0.897	0.930	0.913	0.550	0.940	0.590	0.453	0.538	0.571
-	-	✓	0.899	0.929	0.914	0.548	0.939	0.576	0.427	0.532	0.576
✓	✓	-	0.893	0.922	0.907	0.550	0.932	0.602	0.422	**0.541**	0.568
✓	-	✓	0.896	0.930	0.913	0.542	0.931	0.569	0.431	0.531	0.566
-	✓	✓	0.893	0.922	0.907	0.553	0.934	0.601	0.418	0.539	0.577
✓(1)	✓(1)	✓(1)	**0.911**	0.925	0.918	0.547	0.924	**0.606**	0.446	0.536	0.574
✓(2)	✓(3)	✓(2)	0.904	**0.932**	**0.918**	**0.554**	**0.941**	0.597	**0.481**	0.537	**0.576**

TABLE VII
DIFFERENT CONFIGURATIONS OF DILATION RATES OF DLCM

B→ M → T	P	R	F_1	AP	$AP^{.5}$	$AP^{.75}$	AP^s	AP^m	AP^l
1,1,1	**0.904**	**0.932**	**0.918**	**0.554**	**0.941**	**0.597**	**0.481**	**0.537**	0.576
3,2,1	0.894	0.931	0.912	0.551	0.933	0.593	0.454	0.533	0.576
6,4,2	0.892	0.930	0.911	0.550	0.941	0.596	0.427	0.528	**0.578**

F_1, and AP, obtained by our method decrease continually with the increase of dilation rates from bottom to top feature layers. It might be because of much uninformative background captured by enlarged receptive fields, which causes serious interference on feature extraction for small aircraft. Therefore, setting all dilation rates of deformable convolution within DLCM to 1 is a moderate tradeoff for extracting discrete characteristics of aircraft.

V. CONCLUSION

In this article, an AFRAN is proposed for aircraft detection in SAR images by carefully taking aircraft's domain knowledge and challenges into consideration. Three significant components, including AFFM, DLCM, and ADM, were introduced for adaptively refining and aligning significant discrete features of aircraft. Ablation studies conducted on a self-built aircraft-sliced dataset verified the powerful feature aggregation and refinement abilities of AFFM and effective feature alignment ability of DLCM and ADM by adopting them at the lateral connections and the detection heads of a three-layer fine-grained feature pyramid, respectively. Extensive experiments conducted on the test set of the aircraft sliced dataset and a large scene SAR image demonstrated the topmost detection accuracy as well as competitive speed and spatial complexities achieved by our method in comparison with other domain-specific methods, e.g., DAPN and PADN, and general CNN-based methods, e.g., Faster R-CNN, FPN, Cascade R-CNN, RefineDet, and RPDet. In the future, a further combination between characteristics of aircraft and network design will be investigated to make great promotions for aircraft detection in SAR images.

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